

Ultimate Nutrition® Protein Isolate 2 Scientific Description

During times of physical stress, the body's glutamine reserves become depleted, but can be replenished through glutamic acid supplementation. To address this need, Protein Isolate 2 contains only the highest-grade wheat protein isolate, due to its unique high glutamic acid content. Ultimate Nutrition® Protein Isolate 2 now also includes pea protein isolate, which contains particularly high levels of arginine and lysine, both of which are essential to growth and muscle development.

Weightlifters and bodybuilders realize that glutamine is critical for muscle building because it is a nitrogen donor, meaning that it moves nitrogen around in the body to where it is needed. A positive nitrogen balance is necessary to gain muscle mass. Arginine also plays a role in maintaining nitrogen balance, thereby supporting muscle mass increase and body fat reduction.

Glutamine is the ultimate anti-catabolic. It prevents "muscle wasting" by reducing the degree of muscle damage and soreness caused by heavy exercise. Glutamine also helps the body to secrete its own growth hormone, and subsequently, the release of IGF-1 (Insulin Growth Factor 1). This hormone helps promote weight loss, increase lean body mass, and increase bone density. Glutamine can also neutralize lactic acid burn in muscles by releasing plasma bicarbonate, allowing a longer workout.

Arginine stimulates the release of growth hormone from the anterior pituitary gland, and the release of insulin from the pancreas, which transports nutrients into muscle cells to promote growth while preventing the onset of a detrimental catabolic state.

Lysine stimulates fat mobilization for energy uses, while supporting bone and muscle growth by maintaining healthy calcium absorption. Lysine also assists in recovery and the maintenance of lean body mass by repairing damaged tissues and supporting the body's production of antibodies, hormones, enzymes, and collagen.

Enzymes

Ultimate Nutrition® Protein Isolate 2 includes a blend of three digestive enzymes (protease, bromelain, and papain) to aid in digestion and increase nutrient absorption. These proteolytic enzymes breakdown the protein (macromolecule) into smaller fragments (polypeptides), which are then broken down into smaller peptides and individual amino acids which can then be easily absorbed and utilized by the body.

Due to its broad specificity [wide pH range] bromelain exhibits activity through multiple areas in the digestive tract, including the stomach and small intestine. Additionally, it has been shown to be an adequate replacement for the more common animal-source enzymes such as pepsin and trypsin.

Papain and protease are also very effective in the breakdown of protein. They exhibit both endo- and exopeptidase activity, meaning they break down the macromolecules by internally cleaving them into smaller fragments [chopping it at the middle] and by terminally cleaving the ends of those fragments.

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